FINAL PROJECT DOCUMENTATION

MedTrack Healthcare Dashboard

Hospital Operations & Patient Analytics Dashboard

Module 8 – Documentation and Project Delivery

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1. Project Overview

MedTrack Healthcare Dashboard is a hospital operations and patient analytics project built to turn raw hospital data into interactive, decision-ready visualizations. The project brings together patient admissions, department activity, bed and ICU occupancy, and resource usage into four connected Tableau dashboards, giving hospital stakeholders a single place to monitor how the hospital is performing day to day.

This document covers Module 8 of the project: Documentation and Project Delivery. It consolidates the dataset sources, KPI definitions, dashboard guide, project structure, and deployment plan into a single, submission-ready reference, and briefly confirms that the dashboard suite passed its testing and validation phase (Module 7) prior to this delivery.

2. Project Objectives

The MedTrack project was built with the following objectives:

- Monitor hospital admissions and overall patient load.
- Analyze patient flow, including admission trends, discharge patterns, and length of stay.
- Track bed and ICU occupancy to support capacity planning.
- Monitor readmission patterns across the hospital and by department.
- Compare department-level performance and efficiency.
- Analyze utilization of hospital resources, including beds, staff, and equipment.
- Present all of the above through interactive, filterable Tableau dashboards suitable for non-technical hospital stakeholders.

3. Dataset Sources

The dashboards are built on publicly available synthetic healthcare datasets sourced from Kaggle. Synthetic data was used specifically because complete, real hospital operational datasets are not publicly available due to patient privacy and confidentiality regulations.

Dataset	Description	Source
Hospital Operations Dataset	Hospital-level operational records used to build the prepared and final analytics dataset.	Kaggle (synthetic healthcare data)
HDHI Admission Dataset	Patient admission records contributing to patient flow and clinical fields.	Kaggle (synthetic healthcare data)
Staff Resource Dataset	Staff and staff-schedule records used during data preparation.	Kaggle (synthetic healthcare data)

Exact dataset URLs / Kaggle listing links: To be specified.

4. Dataset Description

The dashboards are powered by the prepared analytics dataset, hospital_final_dataset.xlsx, which combines cleaned hospital operations data with engineered KPI fields. Each row represents a single patient visit. Key fields include:

Field	Description
Patient_ID, Visit_Date, Age, Gender	Patient identifiers and demographics
Disease, Severity, Admission_Type	Clinical classification of the visit
Length_of_Stay, Readmission_Within_30_Days	Stay duration and readmission outcome
Bed_Required, ICU_Required, Oxygen_Required, Ventilator_Required	Resource requirement flags for the visit
Total_Beds_Available, Beds_Occupied, ICU_Beds_Available, ICU_Beds_Occupied	Bed and ICU capacity figures
Oxygen_Units_Used	Oxygen resource consumption
Month, Season, Day_of_Week	Time attributes used for trend analysis
Department	Admitting hospital department

5. Data Preparation / Methodology

The raw hospital datasets were collected, cleaned, and transformed before being used in the dashboards. At a high level, the preparation process followed these stages:

- Data Collection — raw hospital operations, admission, and staff datasets were gathered and consolidated into hospital_raw_data.csv.
- Data Cleaning & Transformation — duplicate records were removed, missing values were handled, department names were standardized, and healthcare indicators were normalized, producing hospital_cleaned.csv.
- KPI Engineering — healthcare KPIs (admissions, occupancy, readmission, length of stay, and related metrics) were calculated from the cleaned dataset, producing the analytics-ready hospital_final_dataset.xlsx used by all four dashboards.

This staged approach keeps the raw, cleaned, and final datasets separate and traceable, so any KPI shown on a dashboard can be traced back to a specific transformation step.

6. KPI Definitions

The dashboard suite is built around six core healthcare KPIs. Each is defined below in plain terms, along with the calculation logic used to compute it from the dataset.

KPI	Definition	Calculation Logic
Total Admissions	The total number of patient visits recorded in the dataset.	Count of distinct Patient_ID values.
Bed Occupancy Rate	The proportion of hospital beds occupied, relative to total beds available.	Sum of Beds_Occupied ÷ Sum of Total_Beds_Available.
ICU Occupancy Rate	The proportion of ICU beds occupied, relative to total ICU beds available.	Sum of ICU_Beds_Occupied ÷ Sum of ICU_Beds_Available.

KPI	Definition	Calculation Logic
Readmission Rate	The percentage of patients readmitted within 30 days of a previous visit.	Count of visits where Readmission_Within_30_Days = “Yes” ÷ Total Admissions.
Average Length of Stay	The average number of days a patient remains admitted.	Average of the Length_of_Stay field.
Patient Load	The average number of patients admitted per day.	Total Admissions ÷ number of distinct Visit_Date values.

These definitions describe the calculation logic used during dashboard development. Exact numerical KPI results are reported separately in the project's QA Checklist and Dashboard Testing Report, produced during Module 7.

7. Dashboard Guide

The MedTrack dashboard suite consists of four dashboards, each focused on a different aspect of hospital operations. Together, they give a complete operational picture, from hospital-wide performance down to department- and resource-level detail.

7.1 Hospital Overview

The Hospital Overview dashboard is the entry point into the dashboard suite. It provides a high-level summary of hospital performance, covering:

- Admissions overview and monthly operational trends
- Bed occupancy monitoring
- Readmission analysis
- Core hospital performance KPIs

It is intended to give a hospital administrator a quick, at-a-glance read on how the hospital is performing before drilling into more specific dashboards.

7.2 Patient Flow

The Patient Flow dashboard focuses on how patients move through the hospital over time. It covers:

- Admission trends and discharge tracking
- Patient movement analysis
- Average length of stay
- Peak patient load monitoring

This dashboard helps identify busy periods, understand how long patients typically stay, and track the balance between admissions and discharges.

7.3 Department Analytics

The Department Analytics dashboard compares hospital departments against one another. It covers:

- Department performance analysis
- Patient volume by department
- Readmission by department

- Department efficiency comparison
- Treatment capacity analysis

It allows evaluators and hospital staff to see which departments are handling the most patients, and how their performance and efficiency compare to one another.

7.4 Resource Utilization

The Resource Utilization dashboard monitors how efficiently hospital resources are being used. It covers:

- Bed utilization analysis
- Staff allocation monitoring
- Equipment utilization tracking
- Capacity planning insights
- Resource availability analysis

This dashboard supports operational and capacity-planning decisions, helping identify where resources are stretched and where capacity remains available.

8. Filters and Dashboard Interactions

All four dashboards share a consistent set of global filters, so the same selection can be applied across the entire dashboard suite:

- Department
- Admission Type
- Gender
- Severity
- Month

Beyond filtering, the dashboards support standard Tableau interactions — selections, tooltips, and highlighting — that let a user explore the data directly rather than reading static charts. Dashboard-to-dashboard navigation allows users to move between Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization without leaving the workbook. These interactions were confirmed to be working correctly during the Module 7 testing and validation phase, and no major issues were identified at that stage.

9. Healthcare Operations Methodology

MedTrack is built around a simple operational logic: hospitals generate large volumes of admissions, occupancy, and resource-usage data every day, and turning that data into visual, filterable insight makes it easier to monitor and plan operations. The dashboard suite supports this in a few concrete ways:

- Patient demand and admission patterns are tracked through admission trends and patient load metrics, helping identify busier periods.
- Hospital occupancy and capacity are monitored through bed and ICU occupancy rates, supporting day-to-day capacity awareness.

- Patient movement and length of stay are tracked to understand how efficiently patients move through the hospital.
- Department performance is compared using volume, readmission, and efficiency metrics, helping identify where attention may be needed.
- Readmission patterns are surfaced both hospital-wide and by department, supporting quality-of-care monitoring.
- Resource utilization and availability are tracked to support planning around beds, staff, and equipment.

In combination, the four dashboards move from a hospital-wide view down to department- and resource-level detail, mirroring how a hospital administrator would naturally narrow down from "how is the hospital doing" to "where exactly is attention needed."

10. Project Structure

The project repository is organized into four top-level folders, separating raw processing logic from data, dashboards, and documentation:

Folder	Purpose	Typical Contents
/scripts	Contains the code used to collect, clean, and transform the hospital data, and to engineer the KPI fields used by the dashboards.	data_collection.py, hospital_cleaning notebook/script, generate_hospital_kpis.py
/data	Contains the datasets at each stage of preparation, from raw to analytics-ready.	hospital_raw_data.csv, hospital_cleaned.csv, hospital_final_dataset.xlsx
/dashboard	Contains the Tableau workbook file(s) that hold the four dashboards.	medtrack_dashboard_v1.twbx, medtrack_dashboard_v2.twbx
/docs	Contains all project documentation, including this document and the Module 7 QA materials.	Final_Documentation_MedTrack_Module8.docx, QA Checklist, Dashboard Testing Report

11. Tools and Technologies

- Tableau — dashboard development and interactive visualization
- Microsoft Excel — final dataset storage and review
- Python — data collection, cleaning, transformation, and KPI engineering
- Visual Studio Code — script development
- GitHub — project version control and repository hosting

12. Project Deliverables

The final MedTrack project delivers the following:

- Four interactive Tableau dashboards (Hospital Overview, Patient Flow, Department Analytics, Resource Utilization)
- The prepared and final healthcare datasets
- Data preparation and KPI engineering scripts

- Complete project documentation, including this Final Documentation, the QA Checklist, and the Dashboard Testing Report
- A GitHub repository organized per the structure defined in Section 10

13. GitHub Repository and Deployment

The project is intended to be delivered as a GitHub repository, organized using the /scripts, /data, /dashboard, and /docs structure described in Section 10, so that an evaluator or portfolio visitor can navigate directly to the code, data, dashboards, or documentation.

14. Tableau Workbook

The four MedTrack dashboards are contained in packaged Tableau workbook files (.twbx), which bundle the dashboards together with the underlying data extract so the workbook can be opened directly in Tableau without a separate data connection step.

Workbook File	Dashboards Included
medtrack_dashboard_v1.twbx	Hospital Overview, Patient Flow
medtrack_dashboard_v2.twbx	Department Analytics, Resource Utilization

15. Conclusion

MedTrack Healthcare Dashboard brings together hospital admissions, patient flow, department performance, and resource utilization data into four connected, interactive Tableau dashboards. The underlying dataset was collected, cleaned, and engineered into a set of core healthcare KPIs, and the resulting dashboards were tested and validated in Module 7, with all documented test cases passing and no major issues identified.

This document completes Module 8 by consolidating the project's documentation — dataset sources, KPI definitions, dashboard guide, project structure, and deployment plan — into a single, portfolio-ready reference, alongside the GitHub repository and Tableau workbook deliverables.

16. Future Enhancements

Potential directions for extending the MedTrack project beyond its current scope include:

- Publishing the workbook to Tableau Public for broader portfolio visibility.
- Connecting the dashboards to a live or regularly refreshed hospital data source, rather than a static dataset.
- Adding predictive analytics, such as forecasting admissions or occupancy trends.
- Expanding department-level analysis with additional operational or financial metrics.
- Adding role-based views (e.g., administrator vs. department head) tailored to different hospital stakeholders.